

Lab Exercises 14 to 16

Date: 21st Oct 2025

Implementation of K-Means Algorithm

from sklearn.cluster import K-Means

1. Apply k-Means clustering on **Wisconsin** (Breast_Cancer) Dataset. Assume $k=2$ and plot a graph for the derived clusters. Also generate elbow graph for this problem and check whether our assumption for k as 2 is true.
2. Dataset: <Housing_Price.csv>
 - (i) Conduct exploratory data analysis on the given dataset and report the details.
 - (ii) Visualize the analysis results using (i) scatter plot (ii) histogram & (iii) box plot.
 - (iii) Implement the K-Means clustering algorithm using the dataset. Try with different k values and show the accuracy.
3. Dataset: <Iris.csv>
 - (i) Conduct exploratory data analysis on the given dataset and report the details.
 - (ii) Visualize the analysis results using (i) scatter plot (ii) histogram & (iii) box plot.
 - (iii) Implement the K-Means clustering algorithm using the dataset. Try with different K values and show the accuracy.